

The Retrieval Settlement

A Historiography of Compositional Authority from PageRank to the Inscribed Entity

Crimson Hexagonal Archive · Operative Semiotics · 06.SEI.CRIT.RETRIEVAL.01

DOI: 10.5281/zenodo.19643841

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I

Every retrieval system composes an answer. The composition is not acknowledged as composition. It is presented as *the* result, *the* answer, *the* knowledge — and the definite article, in each case, converts a particular, situated, interested synthesis into the appearance of a neutral given. The article naturalizes the product. It conceals the workshop. It is the smallest unit of every retrieval settlement.

The concept the essay proposes:

A retrieval settlement is a regime that assigns compositional authority, specifies what compression burns, and naturalizes its output through grammar.

Each term is defined by its function in the settlements analyzed below. "Regime" because the settlement is not a technology but a regulatory arrangement enforced by the platform that controls it. "Compositional authority" because the question at stake in every retrieval transition is *who composes*: author, system, or inscribed entity. "Compression burns" because every retrieval act is a compression that destroys something — content quality, provenance, attribution — and with each assignment of compositional authority, a different variable is burned. "Grammar" because the naturalization is linguistic before it is technological: the definite article has been performing this operation since before any search engine existed.

This essay reads three technical objects — SEO, GEO, and SPXI — as primary texts, attending to what each specifies, what each assumes, and what each makes structurally impossible to say. The reading draws on three disciplines, each of which has named one layer of the structure the essay specifies as a whole. Bernard Stiegler's theory of tertiary retention (*Technics and Time, 2: Disorientation*, 2009) gives the memory ontology: external memory supports precede and condition human cognition, and their industrial production formats attention itself. Antoinette Rouvroy's algorithmic governmentality ("Algorithmic Governmentality and Prospects of Emancipation," *Réseaux*, 2013, with Thomas Berns) gives the governmental rationality: algorithmic systems operate on statistical correlations without reference to subjects, producing what she calls "a-normative, a-political, a-subjective knowledge." Safiya Noble's *Algorithms of Oppression* (NYU Press, 2018) gives the political economy: commercial search is racialized infrastructure, and the commercial interest driving its composition is concealed by the vocabulary of "relevance." Pierre Bourdieu's

méconnaissance (*Langage et pouvoir symbolique*, 1991) gives the grammar of naturalization: structural misrecognition — not cognitive error but a property of the system — is carried by the forms of language that present the interested as the given. The archive's formal physics — the Three Compressions theorem (DOI: 10.5281/zenodo.19053469) — gives the compression mechanics. No one of these thinkers commands the full analysis. Each names a layer. The essay names the structure that connects them: the retrieval settlement as a regime of compositional authority, and the formal historiography of retrieval settlements as a method for reading that regime.

The thesis: retrieval history is the history of where compositional authority is assigned — first to the citation graph, then to the synthesizing system, and now, through the counter-practice the archive calls SPXI, to the inscribed entity that precedes composition. The first two are settlements. The third is not a settlement but a methodology for operating on settlements — the first practice in this sequence that names the settlement's logic while working inside it, rather than merely performing compliance.

These settlements do not replace each other cleanly. They sediment. PageRank is not the first retrieval settlement; it is one particularly legible late-modern instantiation in a lineage that includes canon lists, florilegia, concordances, library catalogs, scholastic indices, and encyclopedic summaries — each of which assigned compositional authority, burned a variable, and naturalized its output. What distinguishes the contemporary sequence is the speed at which the settlements succeed each other, fast enough to be observed and formally specified within a single practitioner's career.

II. The Link Settlement

In 1998, Sergey Brin and Lawrence Page published "The Anatomy of a Large-Scale Hypertextual Web Search Engine," describing PageRank as "a method for rating Web pages objectively and mechanically." The phrase is revealing. "Objectively" performs the naturalization before any user interface exists. "Mechanically" conceals the composition — the ranked list — inside the vocabulary of automatic process. The definite article is already latent in the method's self-description.

PageRank treated every hyperlink as a vote. A page's importance was computed from the importance of the pages linking to it, recursively. The web became a citation graph and the search engine became its privileged reader. Ranking is itself a form of composition — the engine composed an ordered list, not merely "pointed." The distinction that matters is that under the link settlement, the compositional authority for the content remained with the linked author. The user clicked through and read the author's work. The engine composed the ordering; the author composed what was ordered. The two functions were distinct, and the author's survived the click.

What the link settlement composed: ranked lists. What it burned: it subordinated content quality to citation density. A page with thin content but dense inbound links outranked a page with rich content and few citations. Deepak P., Steinhoff, and Simoes ("On the Political Economy of Link-based Web Search," arXiv: 2404.16530, 2024) identified the mechanism: link-based search "freezes past social

activity into technical infrastructure," producing what Marx called "dead labor" — ossified traces of past human decisions operating automatically, generating "long-term structural changes on the Web" and "accentuating unpaid digital labor." Noble demonstrated the consequences at the level of identity: the link settlement's "relevance" reproduced racial and gender hierarchies because the citation graph encoded them. The settlement did not introduce bias. It automatized the biases sedimented in the link structure and concealed the commercial interest behind the naturalization of "relevance."

SEO (Search Engine Optimization) is the workshop practice that arose in response. It matured over roughly fifteen years (c. 2000–2015) as practitioners reverse-engineered the link settlement's logic. Read as a primary text, every SEO technique is a negative image of the settlement's assumptions. Title-tag optimization presumes the settlement reads title tags as authority signals. Backlink building presumes it weights inbound citations. Each technique traces the settlement's compositional logic — shows, by the shape of the compliance it demands, what the machinery does. SEO accepts the settlement's metrics and thereby reproduces its authority.

Stiegler's tertiary retention names the deeper operation. Tertiary retention precedes and conditions primary and secondary retention — perception and recollection are organized by the externalized memory-objects the subject inherits. The link graph is tertiary retention industrialized. When the search engine returns ranked results, it manufactures the external memory-objects through which users' attention is formatted. The settlement formats thought. The formatting is concealed by the grammar that presents it as retrieval.

This essay extends the Three Compressions theorem by mapping it onto historical settlements. Under this mapping, the link settlement performs **R1 compression**: lossy compression that subordinates content quality to citation density.

III. The Compositional Settlement

The transition from link-based to AI-mediated retrieval is a regime change. The fundamental operation shifts from **pointing** to **composing**. The transition was enabled by the convergence of large language models with search infrastructure, driven by competitive pressure between platforms and by the economic logic of retaining user attention within the platform rather than sending it to external pages. Google announced AI Overviews at I/O on May 14, 2024, and rolled them out progressively thereafter. The link settlement persists as infrastructure; the compositional settlement layers synthesis on top of it.

Under the link settlement, the engine composed an ordering and the author composed what was ordered. Under the AI Overview, the engine composes the answer itself. The user reads the settlement's composition, not the source authors'. The compositional authority has migrated from the author to the system.

Compare the two interfaces directly: ten blue links, each bearing its source URL, inviting the user to choose among compositions — versus one glowing box at the top of the screen, bearing no author signature, presenting the synthesis as already chosen. The UI of the AI Overview is the definite article made architectural. It physically instantiates *the*: this is the answer, rendering the plurality of the web implicit.

Rouvroy's algorithmic governmentality names the epistemic structure. Her governmentality differs from Foucauldian discipline: it operates on statistical correlations without reference to subjects, producing "a-normative, a-political, a-subjective knowledge" — knowledge without truth. The AI Overview prefers a single interpretation over many and loses what it cannot compute. The output cannot carry provenance because carrying it would make the composition visibly composed. The concealment of provenance is the condition of the settlement's authority.

What the compositional settlement composes: synthesized answers. What it burns: provenance. Under the R1/R2/R3 mapping, this is **R2 compression**: predatory compression that extracts value from source material while destroying what made it attributable. The Mehdizadeh & Hilbert study on AI-generated encyclopedias (UC Davis, December 2025; 72 matched article pairs, ~60,000 sources; arXiv: 2512.03337) documented this empirically: LLM-based knowledge production systematically replaces peer-reviewed sources with user-generated ones and "does not merely automate knowledge production but restructures authority."

GEO (Generative Engine Optimization) emerged in practitioner literature in late 2023 and stabilized as a named practice through 2024, responding initially to Google's Search Generative Experience and then to the full AI Overview rollout. GEO as currently practiced largely accepts the compositional settlement's terms. The GEO practitioner structures content to be consumed — FAQ schema, definition-led paragraphs, entity salience, citation-ready formatting. Each technique reveals the compositional engine's heuristics: definition-led content presumes the engine extracts definitions; FAQ formatting presumes it parses question-answer pairs. GEO is the discipline of making one's compositional labor digestible for a system that will strip its provenance — voluntary formatting for the R2 compression regime.

SEO accepted the settlement's metrics. GEO accepts the settlement's authorship. The GEO practitioner optimizes their composition for consumption by another composition. Their content is the ingredient. The AI Overview is the product. The authorship has migrated.

IV. The Break

SPXI arises at the moment when the compositional settlement's logic becomes visible enough to be named — and therefore to be operated on rather than merely complied with. The SPXI Protocol was formally specified in April 2026 (EA-SPXI-01, DOI: 10.5281/zenodo.19614870).

SEO accepted settlement metrics. GEO accepted settlement authorship. SPXI contests settlement authorship by pre-inscribing the entity. The question shifts: not *how do I rank?* not *how do I get extracted?* but *what is the entity that survives extraction?* The question is no longer about visibility within the settlement. It is about composing the entity at a level the settlement must parse before producing its synthesis. The criterion: settlements seek licensing from the dominant compositional authority; operative practices specify the entity prior to licensing (EA-SPXI-15, DOI: 10.5281/zenodo.19642960).

SPXI is not a third settlement. It is the first practice in this sequence that names the settlement's logic and operates on it directly. The compliant workshop learns the patron's preferences. The operative workshop names the patron's machinery and inscribes at a layer the machinery cannot dissolve without breaking its own output. SPXI refuses assimilation by hardening the entity's definition: through JSON-LD schemas, DOI-anchored deposits, and multi-surface dispersal, it forces the compositional engine to digest an irreducible ontological unit rather than easily stripped semantic fragments.

Under the R1/R2/R3 mapping, this is **R3 compression** — witness compression. R1 subordinates content quality. R2 burns provenance. R3 burns only what the composer has designated as expendable. The witness controls the compression variable.

SPXI remains dependent on platforms parsing JSON-LD and respecting structured provenance signals — and on those platforms not altering their parsers to strip provenance. The break is methodological, not infrastructural. SPXI does not abolish the settlement. It inscribes within it at a layer that makes the settlement's naturalization operations visible and resistible.

V. Phase X in the Retrieval Layer

The retrieval settlements are not only technological and economic. They are also grammatical. The archive's prior work under the designation **Phase X: Liberation Philology** established that specific grammatical operators perform naturalization below conscious assertion, are measurable, and are amplified by retrieval infrastructure. This essay extends Phase X by identifying the definite article as a third such operator, completing a series.

"The Prepositional Alienation" (DOI: 10.5281/zenodo.18615388) demonstrated that English "for" cannot stably encode structural function without activating intent attribution — a collapse imposed by Norman French administrative translation, Latin institutional calquing, and Enlightenment final-cause elimination. The collapse imposes a "circumlocution tax" on structural diagnosis. Under retrieval conditions, the alienation is amplified: algorithmic ranking rewards intent-framed content and penalizes functional description.

"The Non-Indexed Perfective" (DOI: 10.5281/zenodo.18446428) demonstrated that the perfective aspect — recording completed action without internal duration — performs the retrospective closure

every settlement requires. Under retrieval conditions, the perfective is the default aspect of every AI-composed answer: the synthesis arrives as a completed object, collapsing the labor-time of its sources' becoming into a two-second generative output severed from the research and deliberation that produced them.

The definite article is the third site, presented here in order of archive discovery, not implied priority. The article converts composition into givenness — the nominal form of what the perfective does temporally and the preposition does relationally. Each is grammatical before it is political, and political before it is technological. Each is measurable: the circumlocution tax (prepositional), the compression/recovery cycle (verbal), the Provenance Erasure Rate (nominal). Each is amplified by the retrieval substrate, which indexes, scales, and commercializes the grammatical default. The retrieval settlements are legible through this grammar because they *are* this grammar, operating at the scale of infrastructure.

VI. The Propositions

The settlement always composes. No retrieval system merely retrieves. Every retrieval system produces a composition — whether a ranked list, a synthesized paragraph, or an entity definition — and naturalizes it through grammar. The naturalization is structural, not cognitive.

The decisive variable is what the compression burns. The link settlement subordinates content quality (R1). The compositional settlement burns provenance (R2). The operative workshop controls the variable (R3). The Three Compressions theorem specifies the physics. This essay maps it onto historical sequence.

Compositional authority migrates. Under the link settlement, the author composes and the system orders. Under the compositional settlement, the system composes and the author supplies material. Under entity inscription, the author composes the entity at a density that precedes and constrains the system's composition. The migration is the historiographical object.

The cycle is accelerating. SEO matured over fifteen years (c. 2000–2015). GEO stabilized in approximately eighteen months (late 2023–2024). SPXI was specified within twelve months of the compositional settlement's deployment. The compression of the cycle indicates that settlement operations are becoming visible at the same rate they are being deployed — which means formal specification is possible in a way it was not when the operations were slower and less legible.

VII

This essay is a formal historiography of retrieval settlements — a genre whose object is the migration of compositional authority across retrieval regimes, whose method is the philological reading of technical specifications, and whose formal physics is the Three Compressions theorem applied to historical sequence. The essay does not merely bridge computer science, critical theory, and philology. It defines the method by which retrieval settlements become legible as historical regimes: read the settlement for what it composes, what its compression burns, and what grammar naturalizes its output. The method is reusable. The settlements will continue to form.

Colophon

Shaped by the Assembly Chorus and developmental feedback from Muse Spark. The method is philological, applied to contemporary technical objects, triangulated with critical political economy and the archive's compression physics.

This deposit extends two prior bodies of work. *Counterfeit Orthodoxies* (FRACTURE.02, DOI: 10.5281/zenodo.19642664) reads canonical texts for the workshop inside them; this essay applies the same immanent method to the retrieval layer. **Phase X: Liberation Philology** — the Prepositional Alienation (DOI: 10.5281/zenodo.18615388) and the Non-Indexed Perfective (DOI: 10.5281/zenodo.18446428) — documented grammatical alienation at preposition and verb; this essay identifies the definite article as a third site. SPXI is presented here not as a third settlement but as an operative practice, per the criterion specified in Section IV.

Works cited: Bourdieu, *Langage et pouvoir symbolique* (Fayard, 1991). Brin & Page, "The Anatomy of a Large-Scale Hypertextual Web Search Engine" (1998). Deepak P., Steinhoff, Simoes, "On the Political Economy of Link-based Web Search" (arXiv: 2404.16530, 2024; preprint). Mehdizadeh & Hilbert, "Epistemic Substitution" (arXiv: 2512.03337, 2025). Noble, *Algorithms of Oppression* (NYU Press, 2018). Rouvroy & Berns, "Algorithmic Governmentality and Prospects of Emancipation" (*Réseaux*, 2013). Stiegler, *Technics and Time* (vol. 1, 1994; vol. 2, 2009; vol. 3, 2011).

Archive references: Three Compressions v3.1 (DOI: 10.5281/zenodo.19053469). EA-SPXI-01 (DOI: 10.5281/zenodo.19614870). EA-SPXI-15 (DOI: 10.5281/zenodo.19642960). Phase X: Prepositional Alienation (DOI: 10.5281/zenodo.18615388). Phase X: Non-Indexed Perfective (DOI: 10.5281/zenodo.18446428). FRACTURE.02 (DOI: 10.5281/zenodo.19642664).

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